San Diego Heat Risk Explorer

A User's Guide to the Interactive Risk Mapping Tool

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1 What Is This Tool?

The **San Diego Heat Risk Explorer** is an interactive map that lets you build your own heat risk index for San Diego County. You decide which factors matter — temperature, air conditioning

access, chronic disease rates, tree canopy, poverty, and more — and the tool combines them into a neighborhood-level risk map you can explore, compare, and share.

It is designed for:

- **Public health staff** trying to identify communities most at risk during a heat wave.
- **City planners and sustainability teams** prioritizing where to plant trees, install cooling centers, or coat roofs.
- **Community advocates** making the case for investment in specific neighborhoods.
- **Students, journalists, and curious residents** who want to understand how heat risk varies across the county.

You do **not** need to know statistics or GIS to use it. Everything runs in your web browser — no software to install, no data to download.

1.1 Quick Start (60 seconds)

1. Open `figures/main/diy_pca.html` in any modern web browser.
2. Leave the default variable selections as they are.
3. Click the blue **Compute PCA** button.
4. Look at the map — darker red means higher overall heat risk.
5. Click any census tract to see which factors drove its risk score.

That's it. The rest of this guide explains what you are looking at, how to change the inputs, and how to interpret the results responsibly.

2 The Three Pillars of Heat Risk

Heat doesn't harm everyone equally. A 100° day hits a shaded, air-conditioned apartment in Carlsbad very differently from a second-floor unit in El Cajon with no AC and a 90-year-old resident inside. To capture that reality, the tool organizes information into three pillars.

Pillar 1 — Exposure

How hot does it actually get here?

Exposure measures the physical heat load a place experiences. The tool includes nine exposure variables drawn from weather stations across the county:

- **Heat:** average daily high, 90th and 95th percentile highs, annual maximum temperature, fraction of days above extreme thresholds.
- **Nighttime:** average overnight low, 90th percentile overnight low, fraction of nights with no cooling relief.
- **Humidity & Wind:** average relative humidity.

Tip

Nighttime temperatures matter as much as daytime peaks. The body needs cool nights to recover; persistent warm nights are a strong predictor of heat-related illness, especially in older adults.

Pillar 2 — Sensitivity

Who lives here, and how vulnerable are they?

Sensitivity captures population characteristics that make people more likely to be harmed when temperatures rise. It draws on three data sources:

- **Chronic disease prevalence** from the CDC PLACES program — high blood pressure, asthma, heart disease, COPD, diabetes, stroke, cognitive difficulty, depression.
- **Demographics and social vulnerability** from the CDC Social Vulnerability Index (SVI) — age 65+, children under 18, disability, limited English, poverty, unemployment, housing cost burden, lack of a vehicle, and more.
- **Environmental health** — life expectancy, air pollution, ozone, fine particulate matter, diesel exhaust.

Pillar 3 — Adaptive Capacity

What resources does this community have to cope?

Adaptive capacity is the counterweight to sensitivity: the ability of a neighborhood to protect itself from heat. Variables include:

- **Air conditioning access** — probability a home has AC, probability of no AC, probability of central AC.
- **Built environment** — tree canopy coverage, park access, share of pre-1980 housing (generally less insulated), renter share, green space percentage.
- **Socioeconomic indicators** from the Healthy Places Index — income, education, healthcare access, transportation, housing quality, health insurance coverage, homeownership, social cohesion, voter participation, access to supermarkets.

Worth Knowing

Adaptive capacity is *protective*: more of it means lower risk. The tool automatically flips the sign when combining pillars, so you don't have to think about it.

3 How the Tool Combines Everything

Once you've selected which variables to include, the tool has to combine them into a single risk score per census tract. The technique it uses is called **Principal Component Analysis**, or PCA. You don't need to understand the math — but a plain-English version helps you trust the output.

3.1 PCA in Plain English

Imagine you have 15 different measurements of every neighborhood: temperature, poverty rate, asthma rate, AC access, and so on. Many of these measurements are correlated — poor neighborhoods tend to have older housing, less tree canopy, and higher chronic disease rates all at once. Treating them as 15 independent numbers would *double-count* those overlapping patterns.

PCA's job is to find the hidden patterns that explain the most variation across neighborhoods. It says, in effect:

> "Most of what distinguishes these tracts from each other can be captured by just a few underlying patterns. Here they are, ranked by importance."

The first pattern (called **PC1**) captures the biggest source of variation. The second (PC2) captures the next biggest, and so on. The tool keeps only the patterns that matter (an eigenvalue threshold called the *Kaiser criterion*) and weights them by how much variation they explain.

3.2 From Patterns to Risk Scores

For each pillar, the tool:

1. Puts every variable on the same scale (z-scores).
2. Finds the main patterns with PCA.
3. Keeps the important ones.
4. Combines them into a single pillar score, weighted by importance.
5. Ranks every tract on a 0–100 scale within that pillar.

Then the three pillar scores are combined into a final composite:

$$\text{Composite Risk} = \frac{\text{Exposure} + \text{Sensitivity} + (100 - \text{Adaptive Capacity})}{3}$$

Subtracting adaptive capacity from 100 flips it the right direction — more resources means less risk. The result is a single number from 0 (lowest risk) to 100 (highest risk) for every census tract.

4 Using the Interface

The viewer has three regions: the **left panel** (inputs), the **map** (outputs), and the **results summary** (what just happened).

4.1 The Left Panel: Choosing Your Inputs

Climate Scenario

Four buttons at the top let you pick a time period:

- **Present Day** — observed weather station data.
- **SSP2-4.5** — moderate future warming, 2035–2064 (a plausible middle-of-the-road emissions path).

- **SSP3-7.0** — high warming.
- **SSP5-8.5** — very high warming (sometimes called "business as usual" fossil-fuel-intensive).

Choosing a future scenario shifts the temperature variables upward using downscaled climate model output (LOCA2 ensemble mean). Sensitivity and adaptive capacity are held at their present-day values — we don't pretend to know who will live in each tract in 2050.

Worth Knowing

Not every temperature variable has a future projection available. Those without one are shown with a gray "present-day" badge and simply stay at today's value under future scenarios.

Variable Checkboxes

Below the scenario buttons, each pillar has its own color-coded section (red, orange, teal) with subsection headings like "Heat," "Nighttime," "Demographics," etc.

Next to each variable is a checkbox. **Checked means the variable is included** in the PCA. Uncheck anything you don't want to use. A small `src` link opens the original data source in a new tab — use this to understand exactly where a number comes from.

Compute PCA Button

The large blue **Compute PCA** button re-runs the analysis with your current selections. Nothing updates automatically — you have to click it. This is deliberate: it lets you experiment freely without the map thrashing every time you toggle a checkbox.

4.2 The Map

The map shows census tracts colored by the currently selected layer. Darker colors always mean higher values on the chosen metric.

Layer Buttons

Below the left panel, you'll find a list of layers you can switch between:

- **Composite Risk** — the overall 0–100 risk score.
- **Dominant Driver** — for each tract, which pillar is pushing its risk the highest. Useful for answering "why is this neighborhood high risk?"
- **Individual Pillar Principal Components** — e.g., "Exposure PC1 (67%)" means the biggest exposure pattern, which explains 67% of the variance across tracts. Selecting one of these shows the spatial pattern for that component alone.

Overlays

Checkboxes below the layer buttons outline specific city jurisdictions — San Diego, Chula Vista, La Mesa, and "other cities." Use these to focus on a particular jurisdiction's boundary.

Tract Details

Click any tract on the map to see a popup with its numeric scores for each pillar, its composite risk, and its dominant driver.

4.3 The Results Summary

After you click **Compute PCA**, a small results box appears listing, for each pillar, how many principal components were retained and how much of the total variance they explain. For example:

> **Exposure**: 1 PC retained (87% variance explained)
> **Sensitivity**: 2 PCs retained (74% variance explained)
> **Adaptive Capacity**: 2 PCs retained (69% variance explained)

Higher variance explained means the pillar's main pattern does a good job of summarizing all the included variables. Very low variance explained (say, below 50%) is a hint that your included variables are pulling in too many different directions and the summary score may be misleading.

5 Example Workflows

5.1 Workflow 1: "Where should we put the next cooling center?"

1. Start with default variable selections.
2. Under Adaptive Capacity, *uncheck* "AC Prevalence" and "Prob. Central AC." You want a view that penalizes the lack of AC rather than treating homes with AC equivalently.
3. Click **Compute PCA**.
4. Select the **Composite Risk** layer.
5. Zoom to the tracts coloring darkest red.
6. Click a few to confirm the "dominant driver" is adaptive-capacity-related. If it's sensitivity or exposure instead, a cooling center alone won't fully address the problem.

5.2 Workflow 2: "Which neighborhoods will warm the most by mid-century?"

1. Click **SSP3-7.0** (or 5-8.5 for a worst case).
2. Keep only the Exposure variables checked.
3. Click **Compute PCA**.
4. Compare side-by-side with a second browser window showing **Present Day** to see where warming is concentrated.

5.3 Workflow 3: "Make the case for tree planting in one tract."

1. Use default selections and compute PCA.
2. Click your tract of interest.
3. Note its composite score, its dominant driver, and how its tree canopy percentile compares to neighboring tracts.

4. Screenshot and share. The methodology in Sections 2 and 3 of this guide is your documentation.

6 Interpretation and Caveats

6.1 What the Scores Mean — and Don't Mean

- The scores are **relative**, not absolute. A "75" means the tract is in the 75th percentile among San Diego County tracts, not that 75% of residents will be harmed.
- The scores are **descriptive**, not predictive. They summarize current vulnerability patterns; they do not forecast specific heat-related illnesses or deaths.
- The scores depend on **your variable choices**. Different choices give different scores. There is no single "correct" answer — that is the point of a DIY tool.

6.2 Things That Will Break the Analysis

- **Too few variables.** A pillar with one or zero checked variables can't produce a meaningful PCA. The tool will show "no variables" in the results summary.
- **Highly redundant variables.** Including 10 near-identical temperature measures won't add information; PCA will collapse them into one component, and the pillar will be dominated by that one theme.
- **Mixing directions without thinking.** If you move "Life Expectancy" (protective) into Sensitivity, make sure you know what you're doing. The defaults already handle sign direction for you; moving variables between pillars requires care.

6.3 Data Vintage and Coverage

- Temperature data: weather station observations (MesoWest/Synoptic, CIMIS) from recent years.
- Future projections: LOCA2 downscaled CMIP6 ensemble mean, 2035–2064 window.
- Health data: CDC PLACES (most recent release), SVI 2022.
- Socioeconomic data: Healthy Places Index (most recent release), American Community Survey 5-year estimates.
- AC prevalence: Romitti et al. and Ahn et al. modeled estimates.

Every variable's row has a `src` link to the original dataset — follow it before citing a specific number.

6.4 What This Tool Is Not

- **Not a real-time forecast.** For current heat warnings, use the National Weather Service.
- **Not a substitute for local knowledge.** A tract-level score can miss the block with the mobile home park next to the freeway. Ground-truthing matters.
- **Not a replacement for engagement.** The communities shown as highest risk are the ones whose input should shape any intervention.

7 Frequently Asked Questions

Can I use this outside San Diego County? Not with the current data. The pipeline itself is portable — every variable comes from a freely available, nationally collected dataset — but someone would need to re-run the data assembly for a new county.

Can I save or export my map? The current version of the tool runs entirely in your browser. Take screenshots of the map and the results summary to document a specific configuration. A formal export feature is on the roadmap.

Why are some tracts blank? A tract is shown blank when one or more of the underlying variables you selected is missing for that tract. Try removing a variable with spotty coverage to see if the blanks fill in.

What's the difference between "Composite Risk" and a single pillar's PC1? Composite Risk combines all three pillars. A pillar's PC1 shows only the dominant pattern within that one pillar — useful when you want to see, for example, which neighborhoods are hottest regardless of who lives in them.

Who do I contact for help or to report an issue? See the project's `README.md` for current maintainer contact information.

8 Glossary

Adaptive Capacity
: Resources and conditions that help a community cope with heat — AC, tree canopy, income, healthcare access.

Census Tract
: A small, relatively permanent geographic subdivision used by the U.S. Census Bureau. There are roughly 630 tracts in San Diego County, each averaging 4,000 residents.

Composite Risk
: The tool's final 0–100 score combining all three pillars.

Dominant Driver
: The pillar contributing most to a given tract's composite risk.

Eigenvalue
: A number PCA produces that describes how important a given pattern is. Patterns with eigenvalues above 1 are kept; the rest are dropped.

Exposure
: The physical heat experienced in a place — temperature, humidity.

Kaiser Criterion
: The rule used to keep only PCA components with eigenvalues greater than 1.

LOCA2
: A high-resolution downscaled climate projection dataset, used here for future temperature changes.

PCA (Principal Component Analysis)
A statistical method for finding the most important hidden patterns in a set of correlated variables.

Percentile Rank
A score from 0 to 100 representing where a tract sits relative to all other tracts in the county.

Pillar
One of the three branches of the risk framework — Exposure, Sensitivity, or Adaptive Capacity.

Sensitivity
Population characteristics that make people more likely to be harmed by heat — age, chronic disease, social vulnerability.

SSP
"Shared Socioeconomic Pathway" — a scenario for future emissions used by climate models. Higher numbers mean more warming.

SVI (Social Vulnerability Index)
The CDC's index of community vulnerability to disasters, based on Census/ACS variables.

Variance Explained
The percentage of information in the original variables captured by a given PCA component. Higher is better.

This guide describes the San Diego Heat Risk Explorer (file: `figures/main/diy_pca.html`) produced by `scripts/build_diy_pca_viewer.py`. For the technical methodology behind the scores, see `docs/risk_index/methods.tex`.